

Analysis of Ta-Re Alloys for Use in Superconducting Qubits

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Abstract

Superconducting qubits are among the leading platforms for quantum computing, where material choice critically impacts qubit coherence and scalability. Tantalum (Ta) has emerged as a promising candidate due to its stable, self-limiting oxide (Ta_2O_5), which hosts fewer dielectric two-level systems (TLS). Ta transmons on high-resist silicon exhibit lifetimes (T_1) up to 1.68 ms and Q factors up to 2.5×10^7 [2]. Rhenium (Re) also forms an exceptionally thin and inert oxide layer, and has demonstrated resonator quality factors exceeding 2×10^6 [4]. By alloying Ta with Re, we aim to combine Ta's low-loss coherence performance with Re's enhanced surface and structural properties. Ta-Re alloys may offer tailored superconducting behavior, improved oxidation resistance, and reduced TLS density—potentially surpassing pure Ta in coherence and fabrication resilience. This research explores the stability and oxidation behavior of thin-film Ta-Re alloys deposited on sapphire substrate.

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1 Introduction

1.1 Superconducting Qubits and Decoherence

Superconducting qubits are among the most promising platforms for realizing scalable quantum computers, offering fast gate times, lithographic scalability, and integrated microwave control. Implementations such as the transmon, fluxonium, and flux qubit leverage nonlinear Josephson junctions embedded in superconducting microwave circuits to store and manipulate quantum information. These devices rely on operation at ultra-low temperatures to maintain a superconductive state, minimizing energy loss and preserving quantum coherence.

While state-of-the-art superconducting qubits have achieved remarkable milestones—gate fidelities above 99.9% and integration into multi-qubit processors with quantum error correction techniques [10, 1]—performance remains limited by *decoherence mechanisms* originating at materials interfaces. One of the dominant sources of decoherence is dielectric loss from two-level system (TLS) defects. Existing studies suggest two-level systems arise from structural disorder, trapped charges, and amorphous oxide layers at the substrate-metal, metal-air, and substrate-air interfaces, all of which absorb energy from the qubit’s microwave field [5, 11]. Advancing quantum coherence necessitates the discovery and engineering of new materials that exhibit chemically stable, low-loss surfaces and robust superconducting properties.

1.2 Promising Platforms: Tantalum and Rhenium

Recent research by Chang, Mclellan, et. al. suggests tantalum (Ta) may be a breakthrough platform for superconducting qubits. Tantalum’s advantage lies in its native oxide layer, which forms a ”relatively benign, self-limiting, and stoichiometric Ta_2O_5 film” [11]. This stable passivation reduces the density of amorphous TLS defects compared to traditional materials like niobium or aluminum. Devices fabricated from α -phase Ta (bcc) on high-resistivity silicon substrates have enabled T_1 relaxation times reaching 1.68 ms and resonator quality factors (Q) up to 2.5×10^7 [2].

Similar to Ta, rhenium (Re) has emerged also as a promising candidate for ultra-low-loss superconducting circuits. Re is distinguished by an extraordinarily thin and inert native oxide, which significantly curtails surface dielectric participation in microwave losses. Recent studies on Re-based microwave resonators have demonstrated high internal quality factors exceeding 2×10^6 [4, 6]. Furthermore, transmon qubits leveraging Re as a base material have recently been established as a viable platform for long-lived coherence [16].

However, Re’s utility in monolithic superconducting qubits is constrained by its low bulk critical temperature ($T_c \approx 1.7\text{--}2.0\text{ K}$). Although the low T_c is sufficient for basic cryogenic operation, it heightens sensitivity to thermal quasi-particle generation, limiting standalone deployment.

1.3 Motivation for Tantalum-Rhenium Alloys

To circumvent the limitations of pure Re and further optimize the favorable properties of Ta, material engineering through alloying presents a tempting strategy. Recent work has shown that alloying Ta with other transition metals, such as hafnium (Hf), can systematically tune the superconducting gap and increase T_c while maintaining low-loss characteristics [17]. Similarly, tantalum-rhenium (Ta-Re) alloys offer a highly tunable parameter space aimed at combining the high T_c and process compatibility of Ta with the ultra-thin oxide properties of Re [9]. Ta-Re thin films have further garnered interest in spintronics for their emergent spin Hall conductivity [8], highlighting the rich interplay between their electronic and structural phases.

Of particular interest are Re-rich alloys. Although bulk Re has a low T_c , thin films with substantial Re content may exhibit significantly elevated critical temperatures driven by disorder and morphological modifications. For example, amorphous Re films have shown a $T_c \approx 7\text{--}8\text{ K}$ [14], and electroplated Re in multilayers with noble metals achieves a T_c up to 6 K [12]. An 80% Re and 20% Ta alloy composition seeks to exploit these disorder-induced enhancements in T_c while retaining Re’s promising surface chemistry.

Because surface oxides are the primary host for TLS defects, understanding the exact chemical composition of the Ta-Re oxide interface is critical. Previous X-ray photoelectron spectroscopy (XPS) studies of Re have established its complex oxidation behavior, identifying multiple oxidation states ranging from metallic Re^0 to Re^{7+} in Re_2O_7 , which can vary drastically depending on the environment or the presence of secondary metals like platinum [7, 3, 15]. However, the oxidation dynamics of Re when alloyed with a metal like Ta remain largely unexplored in the context of quantum circuit fabrication.

1.4 Thesis Objectives

The core objective of this thesis is to investigate the surface stability and oxidation behavior of an 80% Re / 20% Ta superconducting thin film to evaluate its viability for low-noise, high-coherence quantum circuits. Using time-resolved X-ray photoelectron spectroscopy (XPS), this systematically profiles the oxide growth and chemical speciation of the Ta-Re alloy following a buffered oxide etch (BOE). By dissecting the resulting surface stoichiometry, this work aims to identify potential sources of TLS dielectric loss and provide insights into engineering optimized Ta-Re interfaces for use in superconducting qubits.

2 Experiment

2.1 Preparation and Measurement

All experiments were done using a previously fabricated 80% Re 20% Ta film deposited on sapphire substrate through co-sputtering. In order to monitor oxide growth, samples were first treated for 20 minutes in 10:1 buffered oxide etch (BOE) solution to etch away most native oxide. While an initial cleanse in piranha solution ($\text{H}_2\text{SO}_4\text{:H}_2\text{O}_2$) was attempted in order to ensure a clean surface, the film on one sample was quickly etched off (see Figure 1). Pure tantalum is known to survive piranha cleaning while pure rhenium does not[16]. One motivation for alloying Ta and Re was to leverage the process compatibility of Ta; however, this result visually demonstrates that a rhenium-rich alloy does not necessarily inherit Ta's chemical robustness to aggressive oxidation, remaining highly susceptible to standard semiconductor acid treatments.

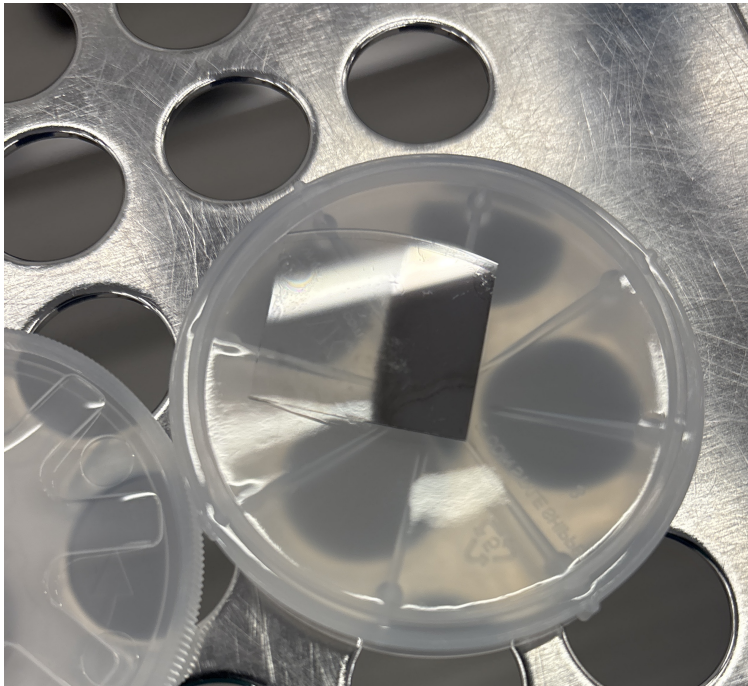


Figure 1: Visual evidence of an 80% Re / 20% Ta thin film completely etched away by a standard piranha solution clean, revealing the underlying sapphire substrate. This indicates that a 20% Ta concentration is insufficient to passivate the alloy against aggressive acid treatments typically survived by pure Ta.

XPS data on Ta and Re was then taken in weekly intervals. Control samples (approximately 6 months of ambient exposure) were kept and measured for comparison. Following XPS collection, scanning electron microscopy data was collected to confirm the structural changes. As shown in Figure 2, even uncorrected spectra showed clear growth.

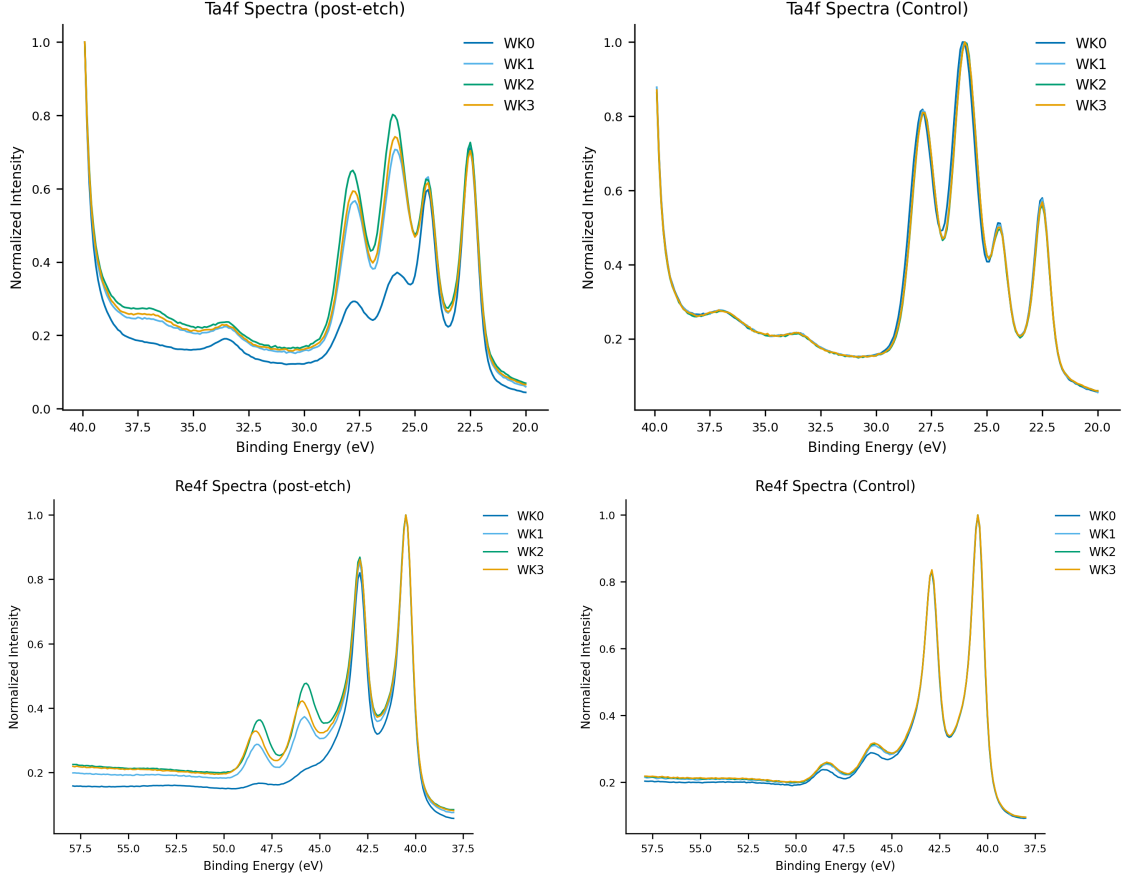


Figure 2: Oxide growth vs. control

2.2 Fitting Procedure

Following Shirley background correction (0.01 cutoff and 0.2 eV step window), XPS intensities were area-normalized and peaks were fitted using an `lmfit` non-linear least-squares minimization procedure [11][7]. Each species was modeled as a doublet attributed to the $4f_{7/2}$ and $4f_{5/2}$ spin-orbit split energy levels.

For Ta4f core electron ionization in the 20 eV to 30 eV range, the model incorporated

six distinct pairs of peaks: a Ta^0 metal bulk state, an interface state, an alloy-shifted state, and three progressive oxidation states (Ta^{1+} , Ta^{3+} , and Ta^{5+} primarily corresponding to Ta_2O_5).

Similarly, for $Re4f$ in the 40 eV to 50 eV range, the model was significantly more complex, encompassing Re^0 metal, three sub-surface oxygen-coordinated states ($Re^{\delta+}$ shifted at +0.22, +0.45, and +0.73 eV above Re^0), and distinct oxide phases corresponding to ReO (Re^{2+}), ReO_2 (Re^{4+}), ReO_3 (Re^{6+}), and Re_2O_7 (Re^{7+}). Detailed fitting constraints for these levels, including spin-orbit splitting and branching ratios, are provided in the Appendix.

To accurately capture the physics of the material surface, asymmetric components (such as metal, alloy, and interface states) were fitted using a Skewed Voigt lineshape to account for conduction electron screening. Conversely, symmetric Gaussian lineshapes were employed for the purely oxide phases.

It should be noted that these fitting parameters were developed primarily for pure Ta and pure Re systems; their applicability to a 80% Re 20% Ta alloy may be limited, and this constraint motivates several of the observations discussed in Section 2.5. Figure 3 shows sample fits for Ta and Re immediately following treatment with BOE. While most Ta data had data vs. fitting residuals < 0.005 , Re saw worse fits with residuals < 0.02 .

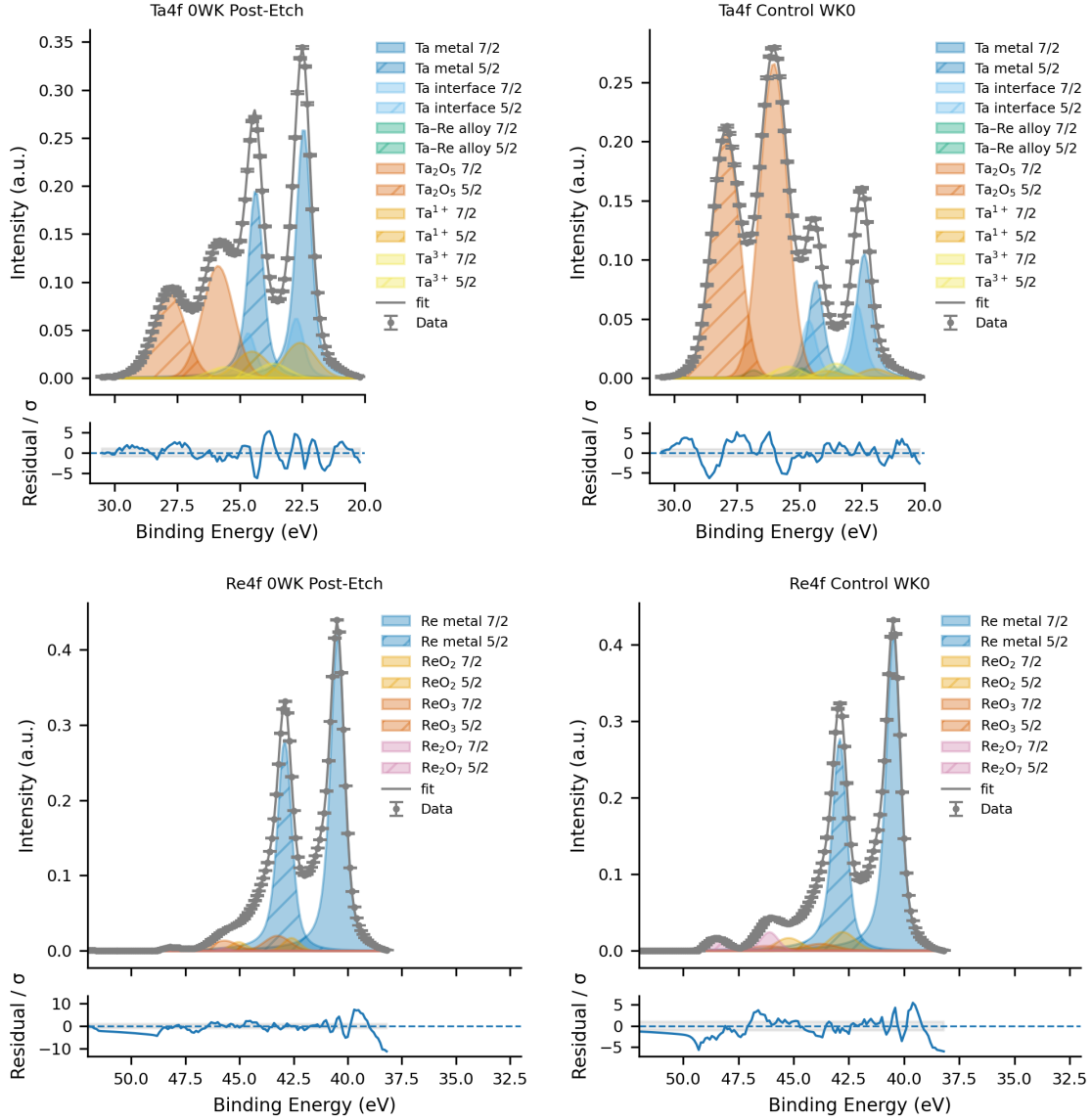
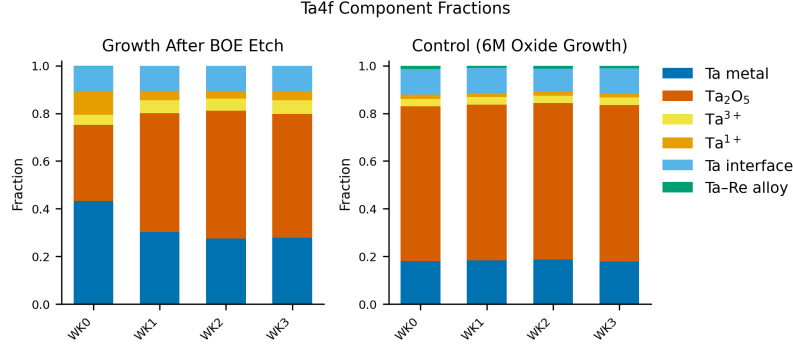
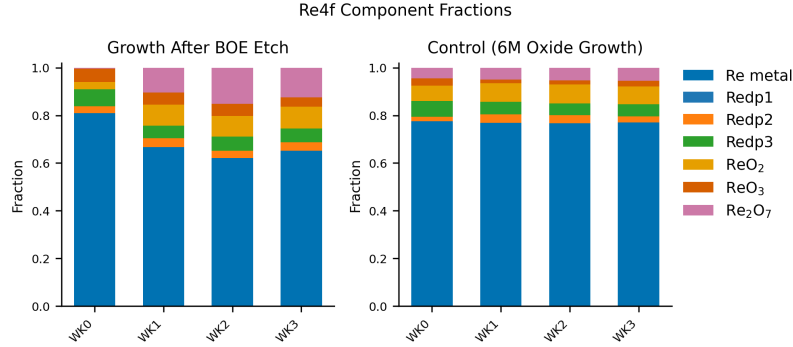


Figure 3: Sample XPS fits

After normalizing data based on the total intensity under the curve, proportions of each species relative to the whole surface were extracted in Figure 4. Similar to pure Ta films, we found Ta_2O_5 grows to become the dominant suboxide species of tantalum. With Re, however, Re_2O_7 seems to see significant growth over the first few weeks but takes up less of the oxide surface after saturation.



(a) Tantalum species



(b) Rhenium species

Figure 4: Tantalum vs. Rhenium oxide species

2.3 Fitting Quality and Constraint Diagnostics

To quantitatively assess the goodness-of-fit across the complex multi-species models, reduced χ^2 statistics were extracted for all scans (Table 1). Overall, the fits maintained stable χ^2 values across the timecourse, though variations indicate the models' differing efficacies in capturing the exact physical lineshapes of BOE-treated versus heavily oxidized control samples.

Sample	Ta4f Reduced χ^2	Re4f Reduced χ^2
BOE_WK0	5.71	7.24
BOE_WK1	6.42	6.29
BOE_WK2	6.27	7.29
BOE_WK3	6.00	6.50
Control_WK0	8.11	4.56
Control_WK1	8.40	5.51
Control_WK2	6.79	5.39
Control_WK3	7.05	4.62

Table 1: Reduced χ^2 goodness-of-fit statistics across all samples.

Additionally, to verify that the large binding energy shifts observed (particularly for the Ta₂O₅ state) were not artefacts of optimizer constraint boundaries, an unconstrained free-fitting diagnostic was performed on the BOE_WK0 sample. After relaxing the Ta⁵⁺ center constraint bounds from [25.5, 28.0] eV to [24.0, 29.0] eV, the fit naturally converged at 25.88 ± 0.01 eV. As seen in Figure 5, this clean convergence nearly 1 eV below the expected pure-Ta position of 26.8 eV confirmed that the observed deviation represents a real initial-state chemical shift rather than a fitting artefact or misidentified sub-oxide state, motivating the deeper investigation of binding energy shifts in Section 2.5.

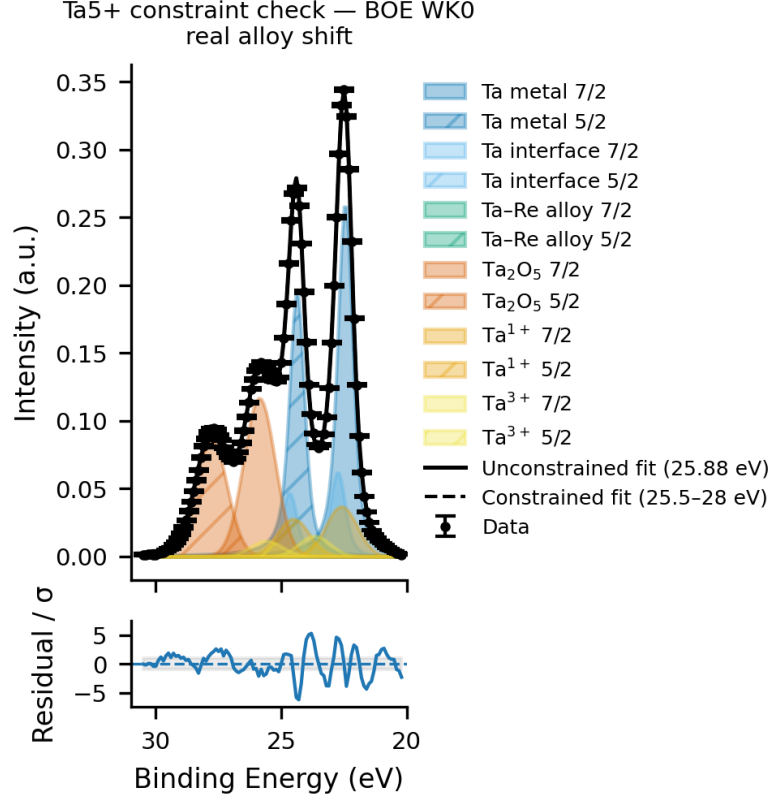


Figure 5: Diagnostic unconstrained fit of the Ta4f core level for the BOE.WK0 sample. The Ta^{5+} peak naturally converges at 25.88 eV, significantly shifted from the standard Ta_2O_5 position (26.8 eV), confirming a genuine chemical shift.

2.4 Thickness Estimation

Following fitting, a very rough estimate of the surface layer was taken using the Strohmeier overlayer model. While more thorough estimation may require variable energy XPS or other methods, Strohmeier assumes the oxide surface is a uniform layer on top of the metal, attenuating the photoelectron signal from the underlying metal based on exponential decay. (It does not account for compositional gradients, surface roughness, etc.[13].)

$$d = \lambda_{\text{ox}} \cos \theta \ln \left[1 + \frac{I_{\text{ox}}}{I_{\text{m}}} \left(\frac{S_{\text{m}}}{S_{\text{ox}}} \right) \left(\frac{N_{\text{m}}}{N_{\text{ox}}} \right) \right] \quad (1)$$

In the expression, d represents the oxide thickness, λ_{ox} the effective inelastic mean free path (IMFP) of photoelectrons in the oxide, and θ the photoelectron take-off angle relative

to the surface. I_{ox} and I_{m} represent the (background-corrected) XPS peak areas associated with the oxide and metallic chemical states, respectively, while the factors S_{ox} and S_{m} are the photoionization sensitivity factors for the corresponding oxide and metal signals. Finally, N_{ox} and N_{m} denote the atomic number densities of the elements.

For this experiment, the X-ray source/collector was normal to the plane while the oxide and metal contributions originated from the same core level of the same element. Under these assumptions, Eq. 1 reduced to the form

$$d \approx \lambda_{\text{ox}} \cos \theta \ln \left[1 + \frac{I_{\text{ox}}}{I_{\text{m}}} \left(\frac{N_{\text{m}}}{N_{\text{ox}}} \right) \right]. \quad (2)$$

Using Eq. 2, the apparent oxide thicknesses associated with both the tantalum and rhenium components were calculated independently. It must be noted here that the Strohmeier model to individual components of an alloy carries significant limitations; the model assumes a uniform, pure oxide overlayer on a pure metal substrate. In a Ta-Re alloy, the surface is a mixed or stratified oxide, meaning Re metal photoelectrons are attenuated by the entire mixed-oxide layer, not just Re oxides. These independent calculations yield “equivalent effective thicknesses” rather than absolute depths of the oxide-metal interface. Yet even with these limitations, Ta dominates the surface oxide network such that the Ta-derived thickness serves as a reasonable proxy for the total oxide layer thickness, while the Re-derived thickness quantifies the relative extent of Re oxidation within that layer.

For the control samples, the equivalent tantalum oxide thickness was consistently around 3.3–3.4 nm, while the equivalent rhenium oxide was very thin, measuring roughly 0.55 nm. Following BOE treatment, the initial equivalent oxide thicknesses (WK0) were substantially reduced to 1.39 nm for Ta and 0.34 nm for Re.

Over the subsequent three weeks of ambient exposure, both oxide species exhibited significant regrowth. To quantify the oxidation kinetics, the time-course thickness data for the

BOE-treated samples were fitted to a Cabrera-Mott growth model:

$$x(t) = x_0 + k \ln \left(1 + \frac{t}{\tau} \right) \quad (3)$$

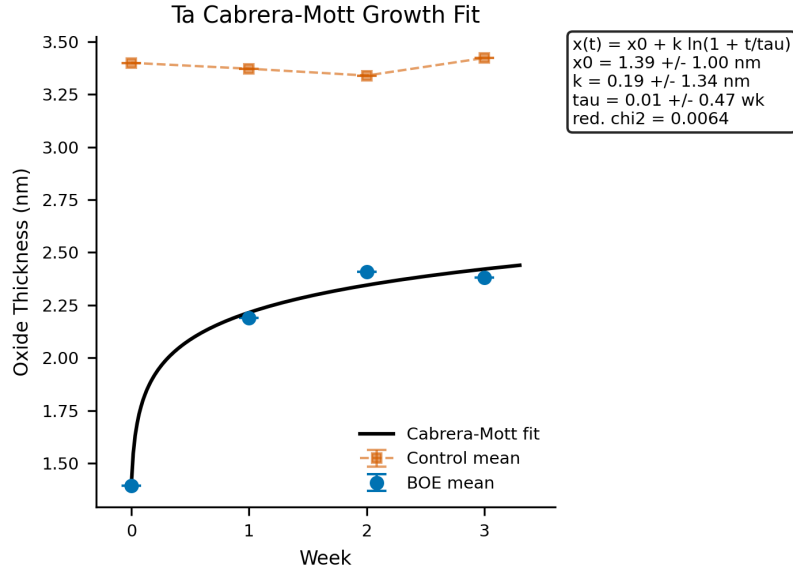
x_0 is the initial thickness, k is the rate constant, and τ is a characteristic time constant.

As shown in Figure 6, the logarithmic model shows rapid initial re-oxidation followed by a slower saturation regime. For tantalum, the oxide regrew to roughly 2.4 nm by week 2, remaining noticeably thinner than the native control oxide (~ 3.4 nm). The Cabrera-Mott fit for Ta yielded $x_0 = 1.39 \pm 1.00$ nm, $k = 0.19 \pm 1.34$ nm, and $\tau = 0.01 \pm 0.47$ weeks. While the uncertainty bounds are broad due to the limited timepoints, the functional form describes the observed stabilization.

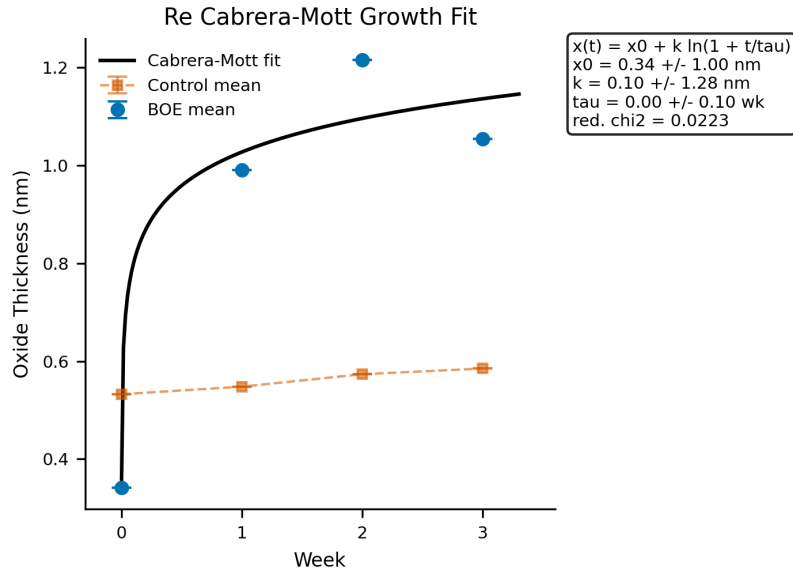
Interestingly, the equivalent rhenium oxide on the BOE-treated samples grew from 0.34 nm to approximately 1.21 nm by week 2 (with fit parameters $x_0 = 0.34 \pm 1.00$ nm, $k = 0.10 \pm 1.28$ nm, and $\tau = 0.001 \pm 0.10$ weeks). These differing growth profiles indicate a highly coupled, two-stage oxidation process within the alloy matrix. In the first stage (rapid oxidation), the removal of the passivating layer exposes both metals to the ambient environment. Lacking immediate passivation, Rhenium oxidizes aggressively. By week 2, the equivalent Re oxide regrows to approximately 1.21 nm—drastically overshooting its stable native thickness on the control sample (~ 0.55 nm).

However, in the second stage (matrix saturation and re-equilibration), the growing Tantalum oxide network saturates at roughly 2.4 nm and begins to effectively passivate the bulk surface [11]. Concurrently, the Rhenium oxide thickness begins to shrink, dropping to 1.05 nm in week 3 and eventually reaching the much thinner state observed in the 6-month control. This kinetic overshoot suggests that once the dominant Ta_2O_5 matrix forms and stabilizes, it halts further Re oxidation. The excess Re oxide that rapidly formed during the unpassivated first stage is subsequently expelled, volatilized, or dissolved from the stabilizing Ta matrix, highlighting that the Ta-Re surface is not a static layer but a system striving

toward a complex chemical equilibrium over several months.



(a) Tantalum Oxide



(b) Rhenium Oxide

Figure 6: Cabrera-Mott logarithmic growth fits for Ta and Re equivalent oxide thicknesses on BOE-treated samples over a 3-week period.

These findings confirm that BOE treatment is effective at nearly eliminating the oxide layer. The speed of growth is also notable; in general, the estimated Ta thicknesses are comparable to accepted values for native tantalum oxides even in an oxide with 80% rhenium

content ($\approx 2.257 \pm 0.023$ nm) [11]. The persistent, evolving mixed-oxide layer is likely a significant source of loss when fabricated into devices.

2.5 Binding Energy and Amplitude Shifts

Beyond thickness variations, the chemical speciation of the surface layer exhibited deviations from pure elemental references. Across both the time-course measurements and long-term control samples, fitted peak centers for various oxidation states shifted consistently, as shown in Figure 7. Metallic Ta 4f components exhibited a large positive binding energy (BE) shift ($\approx +0.42$ eV) relative to pure Ta metal, while metallic Re 4f shifted slightly positive ($\approx +0.10$ eV). Conversely, the primary tantalum oxide (Ta^{5+} in Ta_2O_5) underwent a negative shift (≈ -0.91 eV) compared to its accepted standard position. With Re, the rhenium oxides (Re^{4+} , Re^{6+} , Re^{7+}) exhibited positive shifts ranging from +0.3 to +0.6 eV.

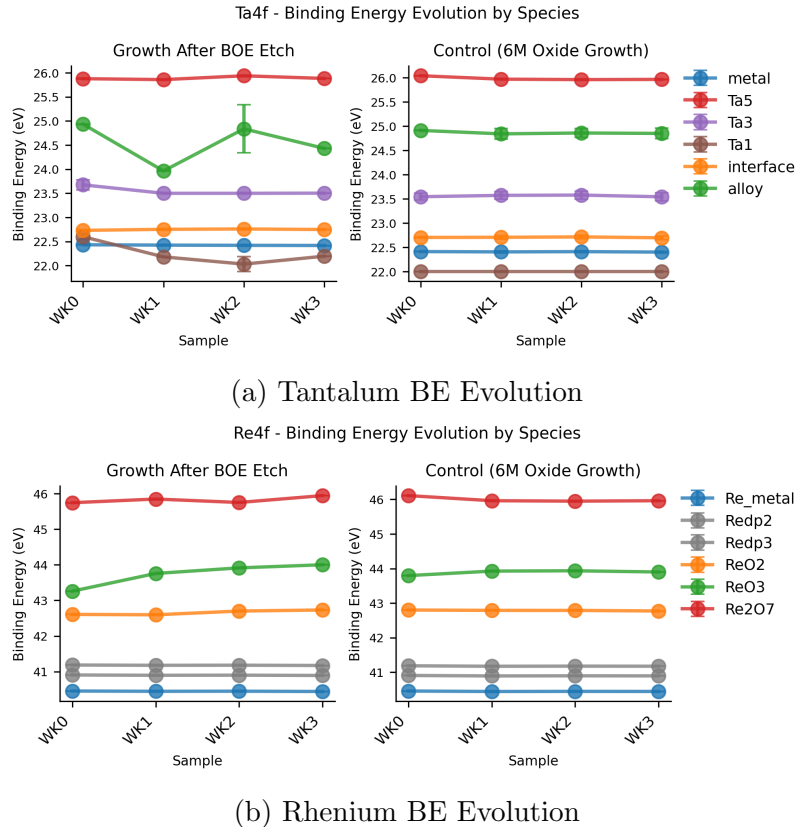


Figure 7: Time evolution of binding energy deviations (Δ eV) from expected pure-element references for Ta and Re species across BOE-treated and control samples.

The opposing nature of these shifts—where oxidized Ta shifts negatively while oxidized Re shifts positively—is summarized in Figure 8. This systematic, opposing movement across the entire sample set strongly points toward a shared electronic environment rather than randomly misidentified independent oxide phases.

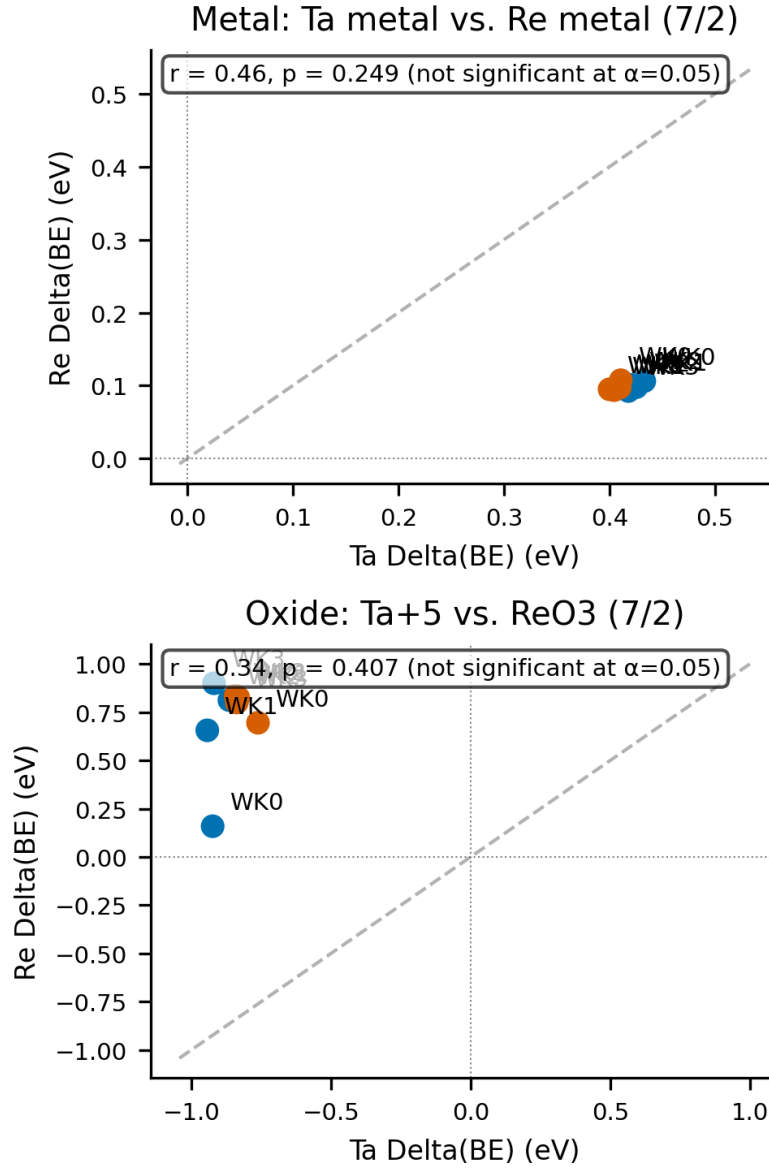


Figure 8: Correlation between Tantalum and Rhenium binding energy shifts relative to standard pure-element references. The opposing shifts (Ta negative, Re positive) are consistently maintained across the growth period and long-term control.

The amplitudes (relative concentrations) of these species also evolved dynamically. Most

notably, as shown in Figure 9, the highly oxidized Re_2O_7 state was nearly non-existent immediately post-etch ($< 1\%$ of the Re signal). Over the next two weeks, it rapidly grew to comprise $\approx 15\%$ of the Re signal. However, in the 6-month control sample, this fraction was significantly reduced to roughly 4–5%. This chemical speciation directly explains the thickness "overshoot" observed in the Re oxide timecourse. The initial burst of unpassivated oxidation heavily favors the formation of high-state Re^{7+} (Re_2O_7). However, Re_2O_7 is known to be highly volatile and hygroscopic, converting readily to perrhenic acid upon exposure to ambient moisture [3, 7]. Thus, as the surrounding Ta_2O_5 matrix saturates and passivates the surface, this unstable, highly oxidized Re^{7+} phase is selectively lost to the environment. The eventual thinning and stabilization of the Re oxide layer is physically driven by the departure of this specific Re_2O_7 component, leaving behind a thinner, lower-oxidation-state Re matrix embedded within the passivating Tantalum oxide.

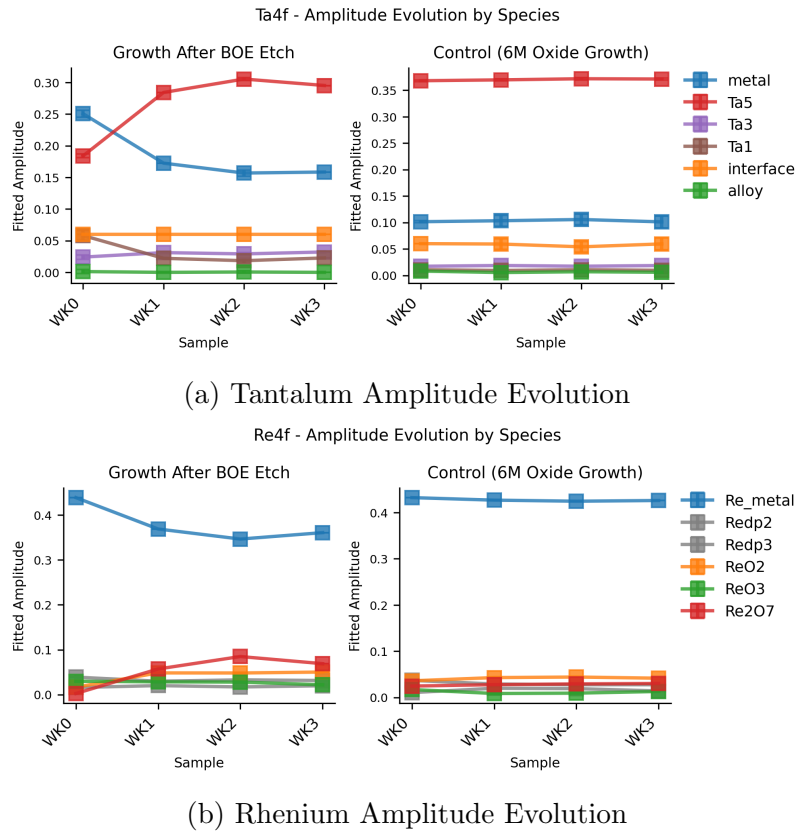


Figure 9: Evolution of normalized XPS peak amplitudes for individual Ta and Re oxidation states across BOE-treated and control samples.

2.6 Further Morphology

Scanning Electron Microscopy (SEM) imaging was performed on a control sample (6 months of ambient exposure) to confirm the sample structure.

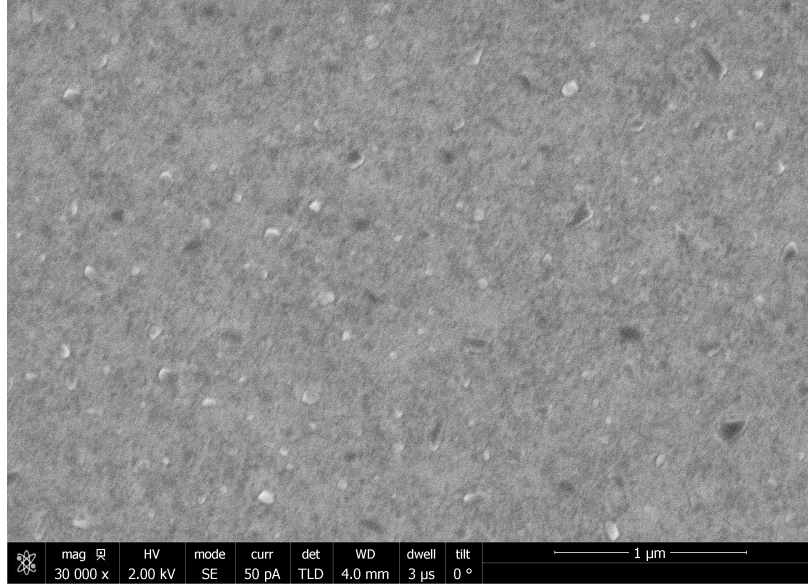


Figure 10: SEM image of the TaRe thin film surface, revealing the surface topography and morphology of the native oxide layer.

Although contamination of sample is apparent in the image, (likely arising from handling and ambient exposure during the exposure period), the underlying surface appears amorphous with botches of uneven morphology.

To further probe the crystalline phases of the oxide layer, an initial grazing incidence X-ray diffraction (GIXRD) analysis was performed on the film. However, the diffractogram was overwhelmingly dominated by the massive reflections of sapphire ($\alpha\text{-Al}_2\text{O}_3$) substrate located at $2\theta \approx 41.6^\circ$ and 90.6° , respectively. Consequently, very few discernible diffraction peaks could be attributed to the TaRe thin film or its native oxides.

While more careful analysis might yield a cleaner signal, the absence of crystalline oxide peaks corroborates the XPS data, indicating that the native oxide layer (Ta_2O_5 and suboxides) is substantially amorphous. This lack of long-range crystalline order within the oxide implies a highly disordered dielectric medium that commonly hosts TLS defects.

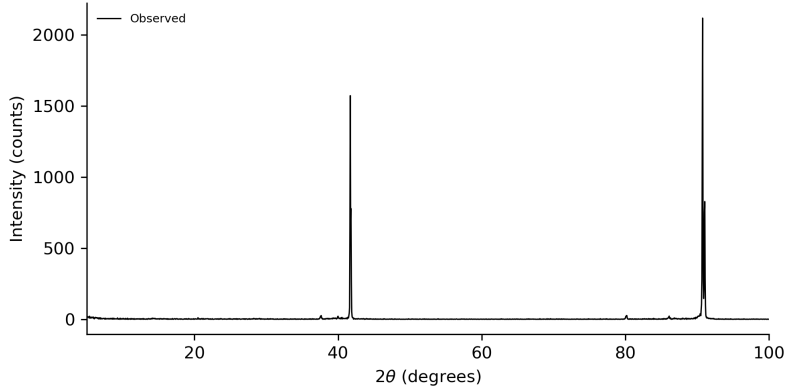


Figure 11: Raw grazing incidence X-ray diffraction (GIXRD) diffractogram of the TaRe thin film, demonstrating the dominance of the sapphire substrate peaks and the lack of distinct crystalline oxide phases.

3 Discussion and Next Steps

This analysis reveals several key features regarding the general growth dynamics of the native oxide on Ta-Re alloys. Most notably, the timecourse data illustrates a coupled, two-stage oxidation process following the initial removal (or "reset") of the surface via Buffered Oxide Etch (BOE). In the first stage, unpassivated rhenium oxidizes aggressively, rapidly forming a thick oxide layer dominated by the Re^{7+} state (Re_2O_7). This leads to a thickness "overshoot" compared to the native equilibrium state shown in the control. In the second stage, the formation of the self-limiting Ta_2O_5 network saturates and passivates the surface [11]. Because Re_2O_7 is highly volatile and hygroscopic [3, 7], it is subsequently dissolved or expelled from the stabilizing matrix. The result is an eventual re-equilibration where the effective rhenium oxide thickness shrinks back down, embedded within a stable Ta-dominant matrix.

This dynamic two-stage growth behavior has significant implications for qubit fabrication. The transient, highly-oxidized state of Re_2O_7 observed during the early growth phase poses a risk as a potential source of TLS dielectric loss; amorphous oxides and fluctuating charge traps associated with such unstable intermediate phases can easily absorb microwave energy,

leading to decoherence [5, 11]. Fabrication workflows for Ta-Re based circuits must either allow sufficient time, employ annealing to let the surface reach thermodynamic equilibrium, or use immediate in-vacuum passivation techniques to prevent the aggressive initial Re oxidation entirely.

The opposing binding energy shifts within the alloy also provide a strong indication of the underlying chemical structure. Because Re is significantly more electronegative than Ta (Pauling scale: 1.9 vs. 1.5), charge transfer naturally occurs from Ta to Re within the 80:20 alloy matrix. This depletion of local electron density around Ta causes its core-level electrons to become more tightly bound, potentially explaining the large +0.42 eV initial-state shift observed for the Ta metal peak. Importantly, the systemic and stable nature of these shifts across both the 3-week growth period and the 6-month control rules out the possibility that these deviations are simply misidentified pure oxide species. If the surface consisted of independent, non-interacting Ta₂O₅ and Rhenium oxide domains, their binding energies would align closely with standard elemental references. Instead, the persistent, opposing shifts may indicate the presence of a distinct ternary Ta-Re-O phase where charge is re-distributed across the mixed lattice.

More detailed analysis using non-destructive techniques like variable-energy XPS and cross-sectional transmission electron microscopy (TEM) is strongly warranted to verify these measurements and accurately construct a depth profile of the alloy surface, especially as standard Ar⁺ ion sputtering was found to induce preferential oxygen reduction rather than cleanly removing the oxide (see Appendix B). All code and data from this analysis are available in the accompanying GitHub repository <https://github.com/dandan002/TaRe-Analysis-Code.git>.

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Appendix A: Fitting Constraints and Tables

Detailed fitting constraints for the Ta4f and Re4f core levels were employed during non-linear least-squares minimization. For Ta4f core electron ionization in the 20 eV to 30 eV range, the spin-orbit splitting offset was constrained near 1.91 eV with a branching ratio of ~ 1.33 . For Re4f in the 40 eV to 50 eV range, the spin-orbit splitting was constrained between 2.35 and 2.50 eV with a ~ 1.33 branching ratio. Asymmetric components were fitted using a Skewed Voigt lineshape to account for conduction electron screening, while symmetric Gaussian lineshapes were employed for purely oxide phases.

Table 2: Fitted binding energies (WK0 - BOE)

Species	Spin	Expected (eV)	Fitted (eV)	StdErr (eV)	Δ (eV)
BOE_WK0					
Ta metal	7/2	22.000	22.433	0.003	0.433
Ta metal	5/2	23.900	24.343	0.003	0.443
Ta interface	7/2	22.400	22.732	0.017	0.332
Ta interface	5/2	24.300	24.642	0.017	0.342
Ta alloy	7/2	24.000	24.933	0.003	0.933
Ta alloy	5/2	25.900	26.843	0.003	0.943
Ta ⁺¹	7/2	22.700	22.600	0.056	-0.100
Ta ⁺¹	5/2	24.600	24.510	0.056	-0.090
Ta ⁺³	7/2	24.200	23.675	0.107	-0.525
Ta ⁺³	5/2	26.100	25.585	0.107	-0.515
Ta ⁺⁵ (Ta ₂ O ₅)	7/2	26.800	25.876	0.010	-0.924
Ta ⁺⁵ (Ta ₂ O ₅)	5/2	28.700	27.786	0.010	-0.914
Re metal	7/2	40.350	40.456	0	0.106
Re metal	5/2	42.770	42.888	0	0.118
Re δ^{+1}	7/2	40.570	40.676	0	0.106
Re δ^{+1}	5/2	42.990	43.108	0	0.118
Re δ^{+2}	7/2	40.800	40.906	0	0.106
Re δ^{+2}	5/2	43.220	43.338	0	0.118
Re δ^{+3}	7/2	41.080	41.186	0	0.106
Re δ^{+3}	5/2	43.500	43.618	0	0.118
ReO (Re ²⁺)	7/2	41.450	41.700	0	0.250
ReO (Re ²⁺)	5/2	43.870	44.132	0	0.262
ReO ₂ (Re ⁴⁺)	7/2	42.200	42.607	0	0.407
ReO ₂ (Re ⁴⁺)	5/2	44.620	45.039	0	0.419
ReO ₃ (Re ⁶⁺)	7/2	43.100	43.259	0	0.159
ReO ₃ (Re ⁶⁺)	5/2	45.520	45.691	0	0.171
Re2O7 (Re7+)	7/2	45.500	45.742	0	0.242
Re2O7 (Re7+)	5/2	47.920	48.174	0	0.254

Table 3: Fitted binding energies (WK0 - Control)

Species	Spin	Expected (eV)	Fitted (eV)	StdErr (eV)	Δ (eV)
Control_WK0					
Ta metal	7/2	22.000	22.411	0.005	0.411
Ta metal	5/2	23.900	24.321	0.005	0.421
Ta interface	7/2	22.400	22.703	0.013	0.303
Ta interface	5/2	24.300	24.613	0.013	0.313
Ta alloy	7/2	24.000	24.911	0.005	0.911
Ta alloy	5/2	25.900	26.821	0.005	0.921
Ta ⁺¹	7/2	22.700	22.000	0	-0.700
Ta ⁺¹	5/2	24.600	23.910	0	-0.690
Ta ⁺³	7/2	24.200	23.543	0.072	-0.657
Ta ⁺³	5/2	26.100	25.453	0.072	-0.647
Ta ⁺⁵ (Ta ₂ O ₅)	7/2	26.800	26.039	0.002	-0.761
Ta ⁺⁵ (Ta ₂ O ₅)	5/2	28.700	27.949	0.002	-0.751
Re metal	7/2	40.350	40.457	0	0.107
Re metal	5/2	42.770	42.884	0	0.114
Re δ^{+1}	7/2	40.570	40.677	0	0.107
Re δ^{+1}	5/2	42.990	43.104	0	0.114
Re δ^{+2}	7/2	40.800	40.907	0	0.107
Re δ^{+2}	5/2	43.220	43.334	0	0.114
Re δ^{+3}	7/2	41.080	41.187	0	0.107
Re δ^{+3}	5/2	43.500	43.614	0	0.114
ReO (Re ²⁺)	7/2	41.450	41.627	0	0.177
ReO (Re ²⁺)	5/2	43.870	44.054	0	0.184
ReO ₂ (Re ⁴⁺)	7/2	42.200	42.800	0	0.600
ReO ₂ (Re ⁴⁺)	5/2	44.620	45.226	0	0.606
ReO ₃ (Re ⁶⁺)	7/2	43.100	43.795	0	0.695
ReO ₃ (Re ⁶⁺)	5/2	45.520	46.222	0	0.702
Re2O7 (Re7+)	7/2	45.500	46.109	0	0.609
Re2O7 (Re7+)	5/2	47.920	48.535	0	0.615

Table 4: Fitted binding energies (WK1 - BOE)

Species	Spin	Expected (eV)	Fitted (eV)	StdErr (eV)	Δ (eV)
BOE_WK1					
Ta metal	7/2	22.000	22.426	0	0.426
Ta metal	5/2	23.900	24.336	0	0.436
Ta interface	7/2	22.400	22.753	0	0.353
Ta interface	5/2	24.300	24.663	0	0.363
Ta alloy	7/2	24.000	23.963	0	-0.037
Ta alloy	5/2	25.900	25.873	0	-0.027
Ta ⁺¹	7/2	22.700	22.182	0	-0.518
Ta ⁺¹	5/2	24.600	24.092	0	-0.508
Ta ⁺³	7/2	24.200	23.500	0	-0.700
Ta ⁺³	5/2	26.100	25.410	0	-0.690
Ta ⁺⁵ (Ta ₂ O ₅)	7/2	26.800	25.857	0	-0.943
Ta ⁺⁵ (Ta ₂ O ₅)	5/2	28.700	27.767	0	-0.933
Re metal	7/2	40.350	40.448	0	0.098
Re metal	5/2	42.770	42.878	0	0.108
Re δ^{+1}	7/2	40.570	40.668	0	0.098
Re δ^{+1}	5/2	42.990	43.098	0	0.108
Re δ^{+2}	7/2	40.800	40.898	0	0.098
Re δ^{+2}	5/2	43.220	43.328	0	0.108
Re δ^{+3}	7/2	41.080	41.178	0	0.098
Re δ^{+3}	5/2	43.500	43.608	0	0.108
ReO (Re ²⁺)	7/2	41.450	41.610	0	0.160
ReO (Re ²⁺)	5/2	43.870	44.040	0	0.170
ReO ₂ (Re ⁴⁺)	7/2	42.200	42.596	0	0.396
ReO ₂ (Re ⁴⁺)	5/2	44.620	45.026	0	0.406
ReO ₃ (Re ⁶⁺)	7/2	43.100	43.756	0	0.656
ReO ₃ (Re ⁶⁺)	5/2	45.520	46.186	0	0.666
Re2O7 (Re7+)	7/2	45.500	45.845	0	0.345
Re2O7 (Re7+)	5/2	47.920	48.275	0	0.355

Table 5: Fitted binding energies (WK1 - Control)

Species	Spin	Expected (eV)	Fitted (eV)	StdErr (eV)	Δ (eV)
Control_WK1					
Ta metal	7/2	22.000	22.405	0.019	0.405
Ta metal	5/2	23.900	24.315	0.019	0.415
Ta interface	7/2	22.400	22.706	0.036	0.306
Ta interface	5/2	24.300	24.616	0.036	0.316
Ta alloy	7/2	24.000	24.840	0.109	0.840
Ta alloy	5/2	25.900	26.750	0.109	0.850
Ta ⁺¹	7/2	22.700	22.000	0	-0.700
Ta ⁺¹	5/2	24.600	23.910	0	-0.690
Ta ⁺³	7/2	24.200	23.571	0.081	-0.629
Ta ⁺³	5/2	26.100	25.481	0.081	-0.619
Ta ⁺⁵ (Ta ₂ O ₅)	7/2	26.800	25.968	0.003	-0.832
Ta ⁺⁵ (Ta ₂ O ₅)	5/2	28.700	27.878	0.003	-0.822
Re metal	7/2	40.350	40.444	0	0.094
Re metal	5/2	42.770	42.874	0	0.104
Re δ^{+1}	7/2	40.570	40.664	0	0.094
Re δ^{+1}	5/2	42.990	43.094	0	0.104
Re δ^{+2}	7/2	40.800	40.894	0	0.094
Re δ^{+2}	5/2	43.220	43.324	0	0.104
Re δ^{+3}	7/2	41.080	41.174	0	0.094
Re δ^{+3}	5/2	43.500	43.604	0	0.104
ReO (Re ²⁺)	7/2	41.450	41.576	0	0.126
ReO (Re ²⁺)	5/2	43.870	44.006	0	0.136
ReO ₂ (Re ⁴⁺)	7/2	42.200	42.790	0	0.590
ReO ₂ (Re ⁴⁺)	5/2	44.620	45.220	0	0.600
ReO ₃ (Re ⁶⁺)	7/2	43.100	43.926	0	0.826
ReO ₃ (Re ⁶⁺)	5/2	45.520	46.356	0	0.836
Re2O7 (Re7+)	7/2	45.500	45.962	0	0.462
Re2O7 (Re7+)	5/2	47.920	48.392	0	0.472

Table 6: Fitted binding energies (WK2 - BOE)

Species	Spin	Expected (eV)	Fitted (eV)	StdErr (eV)	Δ (eV)
BOE_WK2					
Ta metal	7/2	22.000	22.421	0.007	0.421
Ta metal	5/2	23.900	24.331	0.007	0.431
Ta interface	7/2	22.400	22.760	0.011	0.360
Ta interface	5/2	24.300	24.670	0.011	0.370
Ta alloy	7/2	24.000	24.837	0.500	0.837
Ta alloy	5/2	25.900	26.747	0.500	0.847
Ta ⁺¹	7/2	22.700	22.032	0.153	-0.668
Ta ⁺¹	5/2	24.600	23.942	0.153	-0.658
Ta ⁺³	7/2	24.200	23.500	0	-0.700
Ta ⁺³	5/2	26.100	25.410	0	-0.690
Ta ⁺⁵ (Ta ₂ O ₅)	7/2	26.800	25.937	0.004	-0.863
Ta ⁺⁵ (Ta ₂ O ₅)	5/2	28.700	27.847	0.004	-0.853
Re metal	7/2	40.350	40.451	0	0.101
Re metal	5/2	42.770	42.878	0	0.108
Re δ^{+1}	7/2	40.570	40.671	0	0.101
Re δ^{+1}	5/2	42.990	43.098	0	0.108
Re δ^{+2}	7/2	40.800	40.901	0	0.101
Re δ^{+2}	5/2	43.220	43.328	0	0.108
Re δ^{+3}	7/2	41.080	41.181	0	0.101
Re δ^{+3}	5/2	43.500	43.608	0	0.108
ReO (Re ²⁺)	7/2	41.450	41.682	0	0.232
ReO (Re ²⁺)	5/2	43.870	44.109	0	0.239
ReO ₂ (Re ⁴⁺)	7/2	42.200	42.701	0	0.501
ReO ₂ (Re ⁴⁺)	5/2	44.620	45.128	0	0.508
ReO ₃ (Re ⁶⁺)	7/2	43.100	43.913	0	0.813
ReO ₃ (Re ⁶⁺)	5/2	45.520	46.340	0	0.820
Re2O7 (Re7+)	7/2	45.500	45.749	0	0.249
Re2O7 (Re7+)	5/2	47.920	48.176	0	0.256

Table 7: Fitted binding energies (WK2 - Control)

Species	Spin	Expected (eV)	Fitted (eV)	StdErr (eV)	Δ (eV)
Control_WK2					
Ta metal	7/2	22.000	22.410	0.018	0.410
Ta metal	5/2	23.900	24.320	0.018	0.420
Ta interface	7/2	22.400	22.714	0.039	0.314
Ta interface	5/2	24.300	24.624	0.039	0.324
Ta alloy	7/2	24.000	24.857	0.092	0.857
Ta alloy	5/2	25.900	26.767	0.092	0.867
Ta ⁺¹	7/2	22.700	22.000	0	-0.700
Ta ⁺¹	5/2	24.600	23.910	0	-0.690
Ta ⁺³	7/2	24.200	23.576	0.085	-0.624
Ta ⁺³	5/2	26.100	25.486	0.085	-0.614
Ta ⁺⁵ (Ta ₂ O ₅)	7/2	26.800	25.959	0.003	-0.841
Ta ⁺⁵ (Ta ₂ O ₅)	5/2	28.700	27.869	0.003	-0.831
Re metal	7/2	40.350	40.447	0	0.097
Re metal	5/2	42.770	42.876	0	0.106
Re δ^{+1}	7/2	40.570	40.667	0	0.097
Re δ^{+1}	5/2	42.990	43.096	0	0.106
Re δ^{+2}	7/2	40.800	40.897	0	0.097
Re δ^{+2}	5/2	43.220	43.326	0	0.106
Re δ^{+3}	7/2	41.080	41.177	0	0.097
Re δ^{+3}	5/2	43.500	43.606	0	0.106
ReO (Re ²⁺)	7/2	41.450	41.553	0	0.103
ReO (Re ²⁺)	5/2	43.870	43.982	0	0.112
ReO ₂ (Re ⁴⁺)	7/2	42.200	42.789	0	0.589
ReO ₂ (Re ⁴⁺)	5/2	44.620	45.217	0	0.597
ReO ₃ (Re ⁶⁺)	7/2	43.100	43.935	0	0.835
ReO ₃ (Re ⁶⁺)	5/2	45.520	46.364	0	0.844
Re2O7 (Re7+)	7/2	45.500	45.949	0	0.449
Re2O7 (Re7+)	5/2	47.920	48.378	0	0.458

Table 8: Fitted binding energies (WK3 - BOE)

Species	Spin	Expected (eV)	Fitted (eV)	StdErr (eV)	Δ (eV)
BOE_WK3					
Ta metal	7/2	22.000	22.418	0	0.418
Ta metal	5/2	23.900	24.328	0	0.428
Ta interface	7/2	22.400	22.749	0	0.349
Ta interface	5/2	24.300	24.659	0	0.359
Ta alloy	7/2	24.000	24.431	0	0.431
Ta alloy	5/2	25.900	26.341	0	0.441
Ta ⁺¹	7/2	22.700	22.197	0	-0.503
Ta ⁺¹	5/2	24.600	24.107	0	-0.493
Ta ⁺³	7/2	24.200	23.503	0	-0.697
Ta ⁺³	5/2	26.100	25.413	0	-0.687
Ta ⁺⁵ (Ta ₂ O ₅)	7/2	26.800	25.881	0	-0.919
Ta ⁺⁵ (Ta ₂ O ₅)	5/2	28.700	27.791	0	-0.909
Re metal	7/2	40.350	40.443	0	0.093
Re metal	5/2	42.770	42.868	0	0.098
Re δ^{+1}	7/2	40.570	40.663	0	0.093
Re δ^{+1}	5/2	42.990	43.088	0	0.098
Re δ^{+2}	7/2	40.800	40.893	0	0.093
Re δ^{+2}	5/2	43.220	43.318	0	0.098
Re δ^{+3}	7/2	41.080	41.173	0	0.093
Re δ^{+3}	5/2	43.500	43.598	0	0.098
ReO (Re ²⁺)	7/2	41.450	41.648	0	0.198
ReO (Re ²⁺)	5/2	43.870	44.073	0	0.203
ReO ₂ (Re ⁴⁺)	7/2	42.200	42.731	0	0.531
ReO ₂ (Re ⁴⁺)	5/2	44.620	45.156	0	0.536
ReO ₃ (Re ⁶⁺)	7/2	43.100	44.000	0	0.900
ReO ₃ (Re ⁶⁺)	5/2	45.520	46.426	0	0.906
Re2O7 (Re7+)	7/2	45.500	45.941	0	0.441
Re2O7 (Re7+)	5/2	47.920	48.367	0	0.447

Table 9: Fitted binding energies (WK3 - Control)

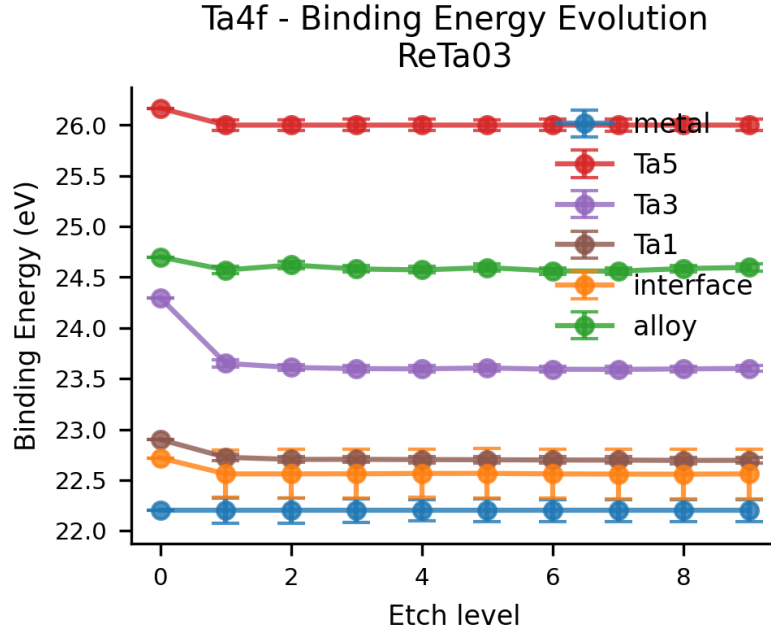
Species	Spin	Expected (eV)	Fitted (eV)	StdErr (eV)	Δ (eV)
Control_WK3					
Ta metal	7/2	22.000	22.400	0.020	0.400
Ta metal	5/2	23.900	24.310	0.020	0.410
Ta interface	7/2	22.400	22.695	0.037	0.295
Ta interface	5/2	24.300	24.605	0.037	0.305
Ta alloy	7/2	24.000	24.848	0.107	0.848
Ta alloy	5/2	25.900	26.758	0.107	0.858
Ta ⁺¹	7/2	22.700	22.000	0	-0.700
Ta ⁺¹	5/2	24.600	23.910	0	-0.690
Ta ⁺³	7/2	24.200	23.541	0.082	-0.659
Ta ⁺³	5/2	26.100	25.451	0.082	-0.649
Ta ⁺⁵ (Ta ₂ O ₅)	7/2	26.800	25.963	0.003	-0.837
Ta ⁺⁵ (Ta ₂ O ₅)	5/2	28.700	27.873	0.003	-0.827
Re metal	7/2	40.350	40.445	0	0.095
Re metal	5/2	42.770	42.871	0	0.101
Re δ^{+1}	7/2	40.570	40.665	0	0.095
Re δ^{+1}	5/2	42.990	43.091	0	0.101
Re δ^{+2}	7/2	40.800	40.895	0	0.095
Re δ^{+2}	5/2	43.220	43.321	0	0.101
Re δ^{+3}	7/2	41.080	41.175	0	0.095
Re δ^{+3}	5/2	43.500	43.601	0	0.101
ReO (Re ²⁺)	7/2	41.450	41.533	0	0.083
ReO (Re ²⁺)	5/2	43.870	43.960	0	0.090
ReO ₂ (Re ⁴⁺)	7/2	42.200	42.772	0	0.572
ReO ₂ (Re ⁴⁺)	5/2	44.620	45.199	0	0.579
ReO ₃ (Re ⁶⁺)	7/2	43.100	43.901	0	0.801
ReO ₃ (Re ⁶⁺)	5/2	45.520	46.327	0	0.807
Re2O7 (Re7+)	7/2	45.500	45.962	0	0.462
Re2O7 (Re7+)	5/2	47.920	48.388	0	0.468

Appendix B: Ar⁺ Ion Depth Profiling and Preferential Sputtering

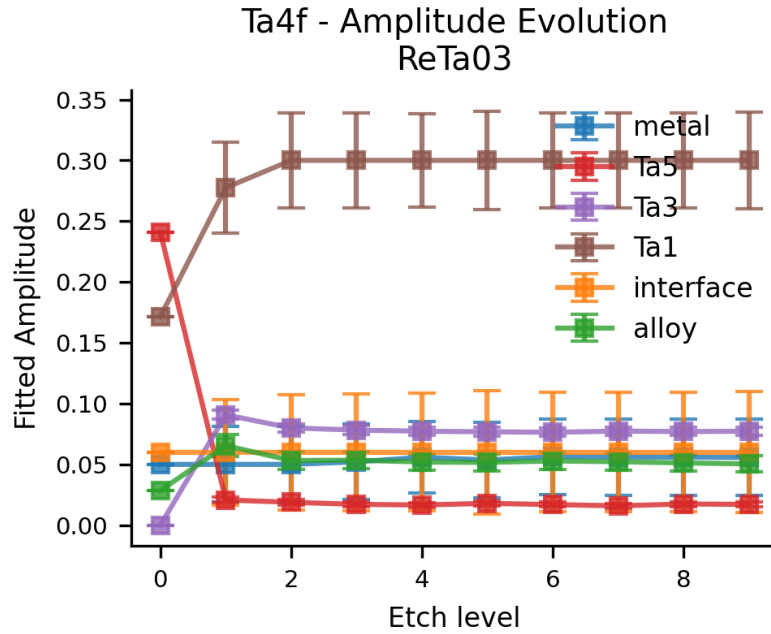
In order to directly measure the thickness and chemical stratification of the native oxide layer, an XPS depth profile using monoatomic Ar⁺ ion sputtering was attempted. However, the results presented ion-beam-induced reduction.

As shown in Figure 12, the area fractions of the highest oxidation states (Ta⁵⁺ in Ta₂O₅ and Re⁷⁺ in Re₂O₇) plummet to near-zero immediately after the first 10 seconds of Ar⁺ etching. If the beam were etching through a very thin oxide layer to reveal the bulk alloy beneath, this drop in oxide signal would be accompanied by a spike in the metal fraction (approaching 100%). Instead, the metallic Ta fraction remains flat at roughly 9% throughout the entire etching process. Rather than exposing the metal, the lost Ta⁵⁺ signal shows artificially inflated sub-oxides, with Ta¹⁺ jumping to $\approx 49\%$ and Ta³⁺ jumping to $\approx 16\%$. A similar reduction occurs in the Rhenium spectra.

The estimated physical thickness of the oxide layer (calculated via the Strohmeier) also barely decreases during the sputtering process. This indicates that the heavy Ar⁺ ions preferentially knocked the lighter oxygen atoms out of the oxide lattice faster than they removed the heavy metal atoms. Because the beam destroyed the chemical state of the surface instead of cleanly milling through it, the resulting depth profile reflects preferential oxygen sputtering artifacts rather than the true native oxide structure. This necessitates the future use of non-destructive depth profiling methods.



(a) Tantalum Species Fraction vs. Etch Time



(b) Rhenium Species Fraction vs. Etch Time

Figure 12: Evolution of XPS area fractions during Ar⁺ ion sputtering. The immediate conversion of high oxidation states (Ta⁵⁺) into intermediate sub-oxides (Ta¹⁺, Ta³⁺) without a corresponding recovery of the metal fraction demonstrates preferential oxygen sputtering.

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